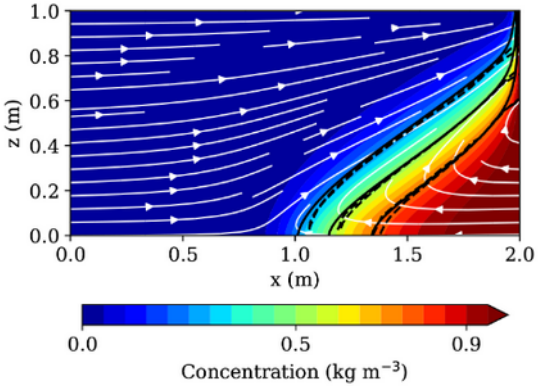
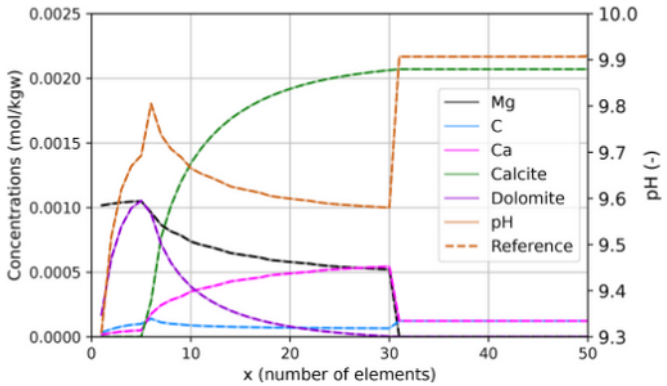
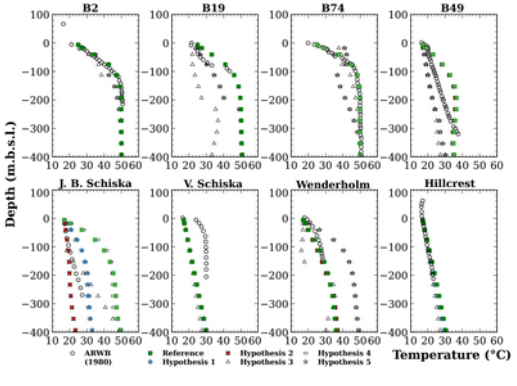
TRANSPORT Simulation Environment

Introduction

Thomas Kempka
GFZ German Research Centre for Geosciences
University of Potsdam







Why Yet Another Numerical Simulator (YANS), not even providing the features of existing ones?

- We have extensively tested {put any transport simulator here} and are not satisfied with the capabilities of any of these
- We do not want to spend multiple months to understand simulators other people have developed (we have been doing this over the past 14 years :-/)
- We want to be flexible and couple/integrate processes as required (T, H, C, M?!, R, B, E)
- We want to understand what we are doing and especially any potential limitations, so that we do NOT leave our scientific work to black boxes
- We have decided that we would like to abandon complex grids, but rather focus on processes and their coupling (→ FDM suffices)
- Most of us are really tired of developing workarounds to get coupled process simulations working, producing questionable results
- We need a simple tool that WE are able to understand, use and adapt

TRANSPORTSE (YANS) benefits in a nutshell...

- FDM implementation of coupled density-driven Darcy flow, heat and species transport < 1500 lines(!) of Python code
- Explicit, semi-implicit and fully implicit solution of coupled equations (simple iterative SOR Gauss-Seidel solver)
- Just-in-time (JIT) compilation (numba) provides execution times in the same order of C/C++/FORTRAN codes
- EOS implementation is straight forward (parallelisation feasible)
- Chemical module coupling straight forward (parallelisation feasible)
- Runs in parallel on multiple CPU cores already scales with 2 cores on your lousy ;-) laptop (tested with 80 cores on GFZ cluster)
- Very efficient and simple model parametrisation thanks to numpy
- Generally very good Python code readability supports understanding of TRANSPORTSE implementation and eases modifications
- HOWEVER, crap in → crap out: you are the modeller and responsible for your work

TRANSPORTSE solves coupled systems of equations for density-driven Darcy flow, species and heat transport

Density-driven Darcy flow (Pressure formulation)

$$\rho_f\left((1-\phi)\alpha+\phi\beta\right)\frac{\partial P}{\partial t}=\nabla\left[\frac{k\rho_f}{\mu_f}\left(\nabla P-\rho_f g\nabla z\right)\right]+W$$

Diffusive and advective species transport (one PDE system for each species)

$$\phi\frac{\partial C}{\partial t}=\nabla(\phi D\nabla C-qC)+Q$$

Conductive and convective heat transfer

$$\left((1-\phi)c_r\rho_r+\phi c_f\rho_f\right)\frac{\partial T}{\partial t}=\nabla\left[\left((1-\phi)\lambda_r+\phi\lambda_f\right)\nabla T-qc_f\rho_f T\right]+H$$

Non-linearity due to parameter dependency on pressure, temperature and dissolved species/single-phase fluid composition and chemical reactions (R)

$$\rho_f(P,T,C),\beta(P,T,C),\mu_f(P,T,C),\phi(R),k(\phi),c_r(T),c_f(P,T,C),\lambda_r(T),\lambda_f(P,T,C)$$

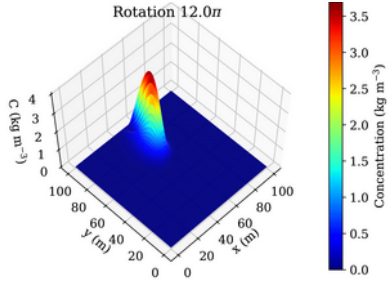
Generic model verification based on density-driven flow

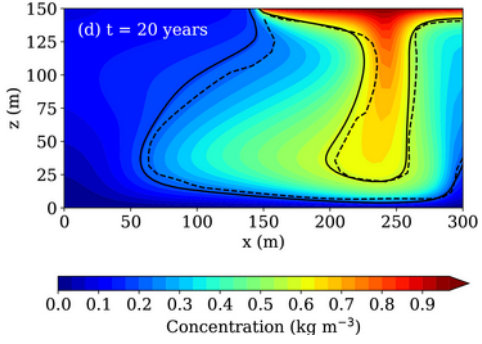
Adv. Geosci., 54, 67–77, 2020
https://doi.org/10.5194/adgeo-54-67-2020
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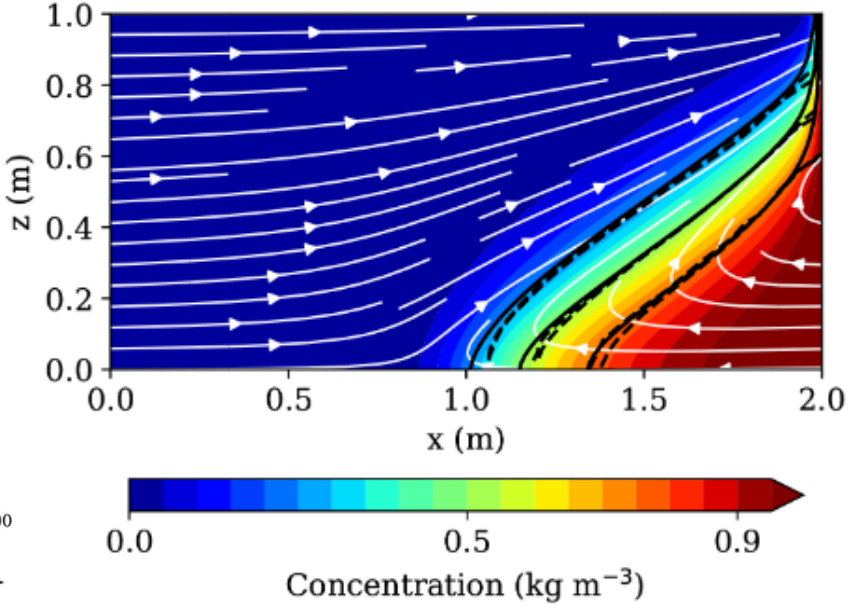
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Verification of a Python-based TRANsport Simulation Environment for density-driven fluid flow and coupled transport of heat and chemical species

Thomas Kempka[1,2]





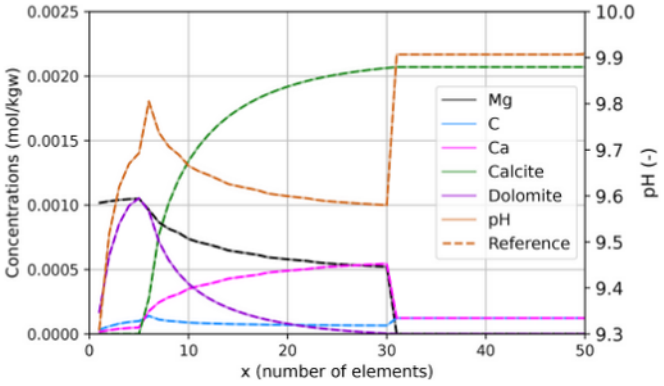


https://adgeo.copernicus.org/articles/54/67/2020/

TRANSPORTSE-PHREEQC coupling verification

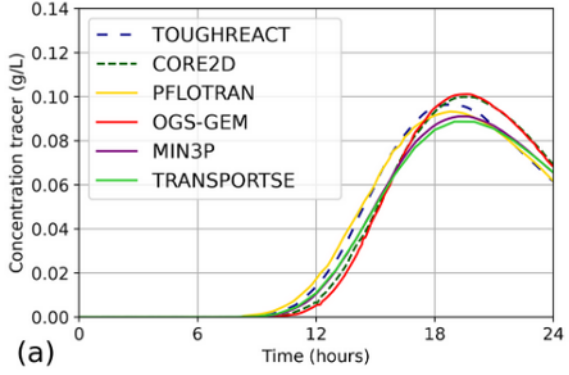
Adv. Geosci., 58, 19–29, 2022
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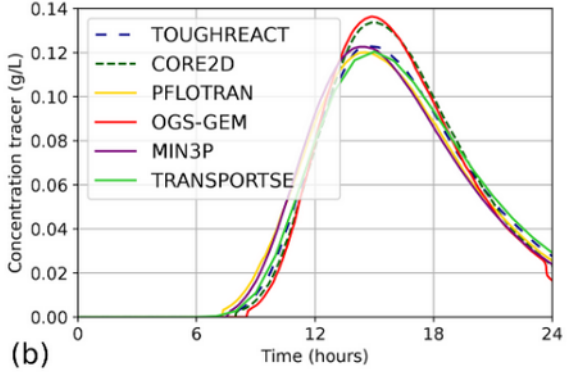
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Verification of TRANSPORT Simulation Environment coupling with PHREEQC for reactive transport modelling

Thomas Kempka[1,2], Svenja Steding[1], and Michael Kühn[1,2]





High density changes as well as solid salt dissolution and precipitation related to potash salt mining in Germany

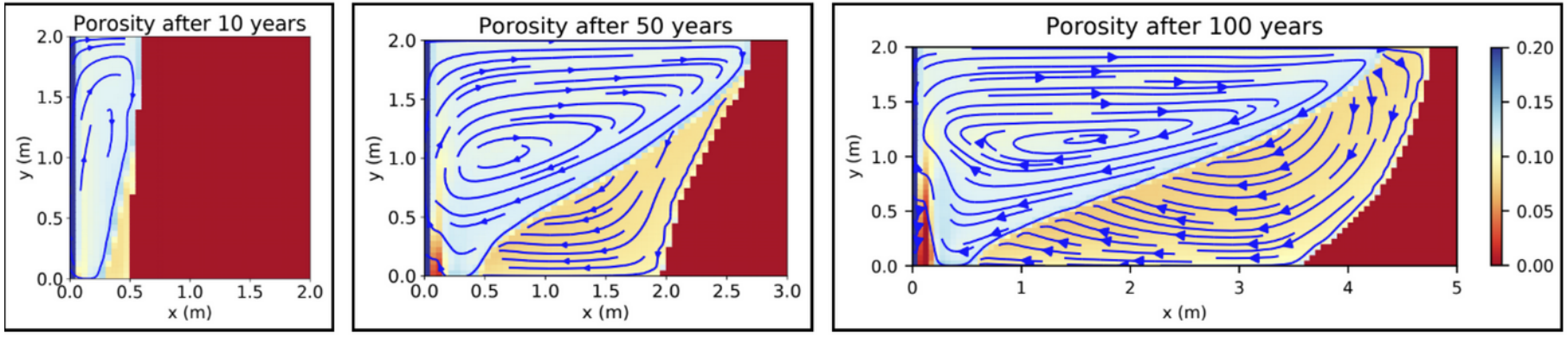
water

MDPI

Article

Spatial and Temporal Evolution of Leaching Zones within Potash Seams Reproduced by Reactive Transport Simulations

Svenja Steding [1,2,*], Thomas Kempka [1,2], Axel Zirkler [3] and Michael Kühn [1,2]



Understanding the dynamics of salt dissolution in heterogeneous geological systems

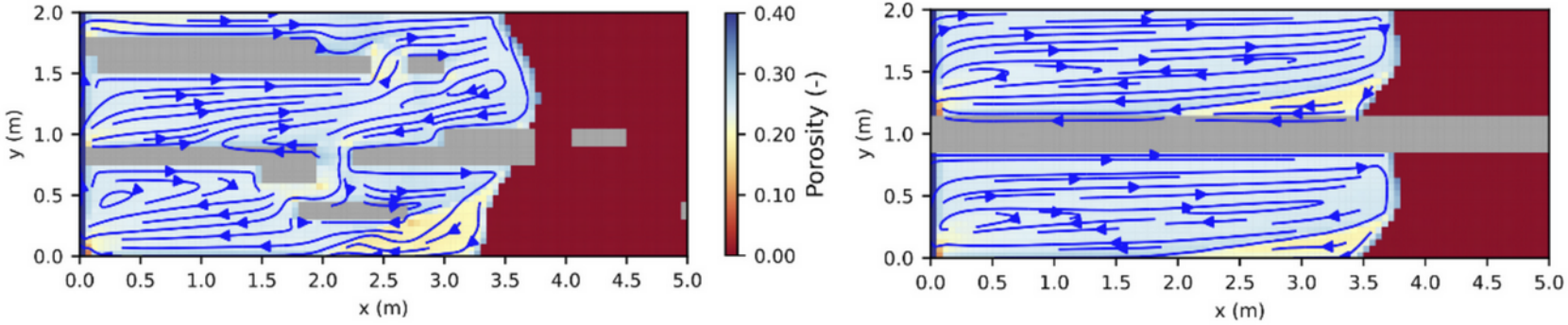
applied sciences

MDPI

Article

How Insoluble Inclusions and Intersecting Layers Affect the Leaching Process within Potash Seams

Svenja Steding [1,2,*], Thomas Kempka [1,2] and Michael Kühn [1,2]



Gas hydrate formation in laboratory-scale reactor

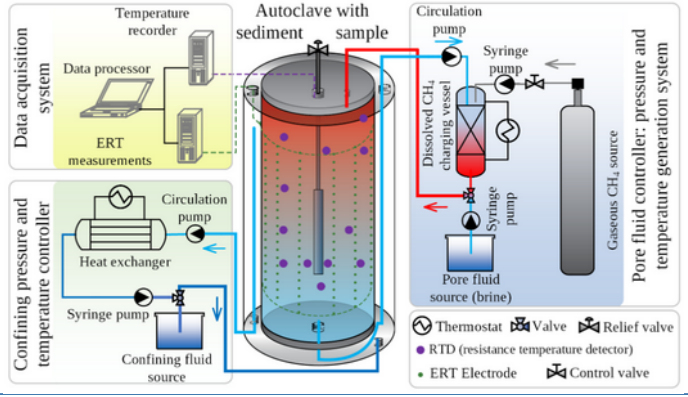
energies

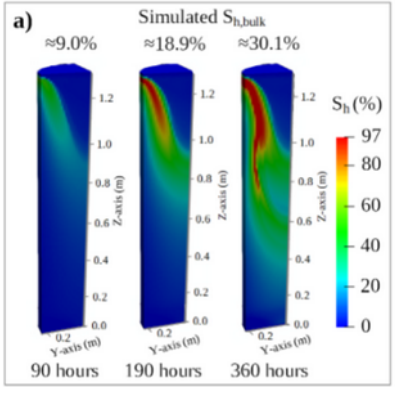
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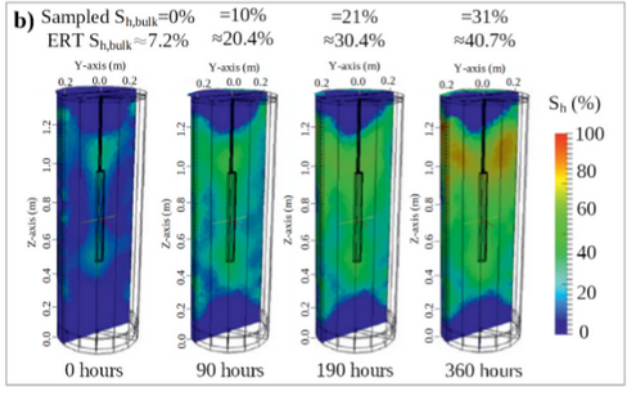
Article

Numerical Simulation of Hydrate Formation in the LArge-Scale Reservoir Simulator (LARS)

Zhen Li [1,2], Erik Spangenberg [1], Judith M. Schicks [1,3,*] and Thomas Kempka [1,2]







Permafrost freezing and thawing in view of methane gas hydrate formation in Mackenzie Delta

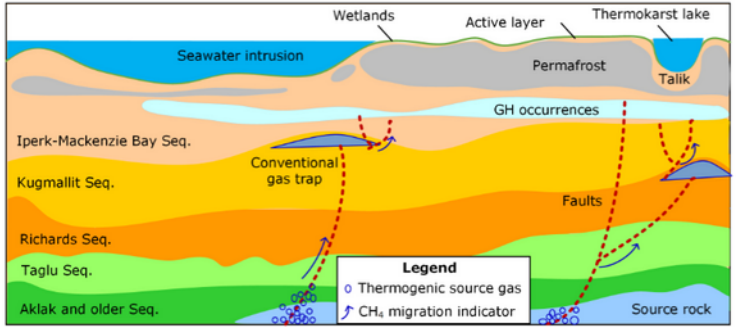
energies

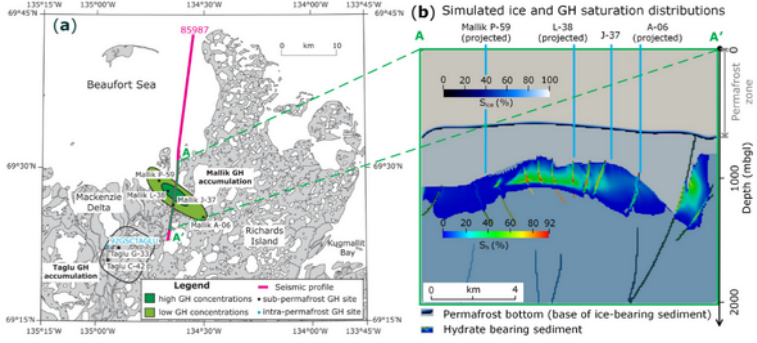
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Article

Numerical Simulation of Coastal Sub-Permafrost Gas Hydrate Formation in the Mackenzie Delta, Canadian Arctic

Zhen Li [1,2], Erik Spangenberg [1], Judith M. Schicks [1,3] and Thomas Kempka [1,2,*]





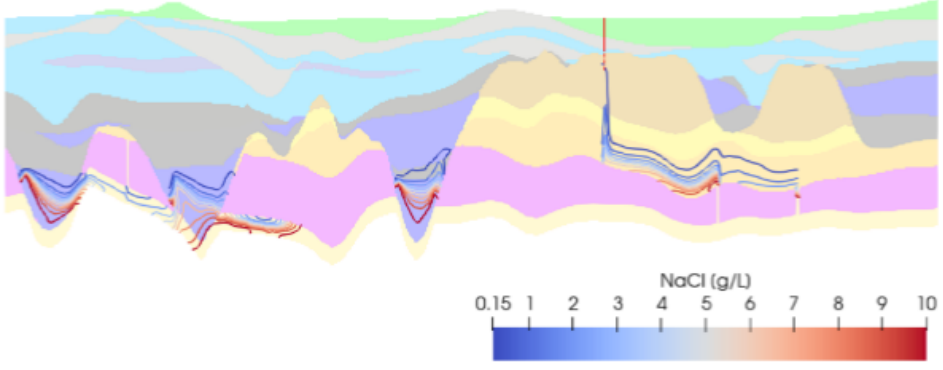
Groundwater salinisation resulting from freshwater production and climate change in Brandenburg

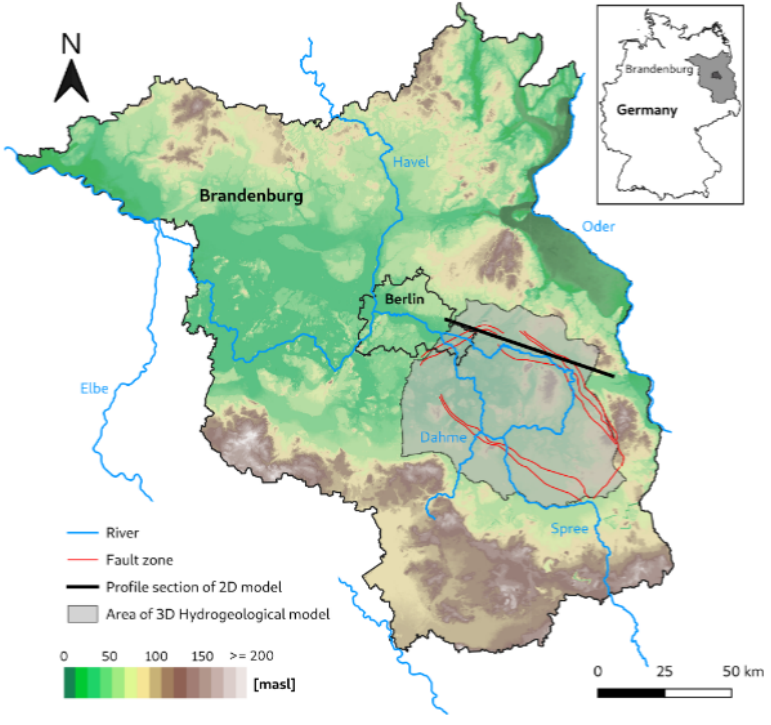
Adv. Geosci., 58, 47–54, 2022
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Upwelling mechanisms of deep saline waters via Quaternary erosion windows considering varying hydrogeological boundary conditions

Elena Chabab[1], Michael Kühn[1,2], and Thomas Kempka[1,2]





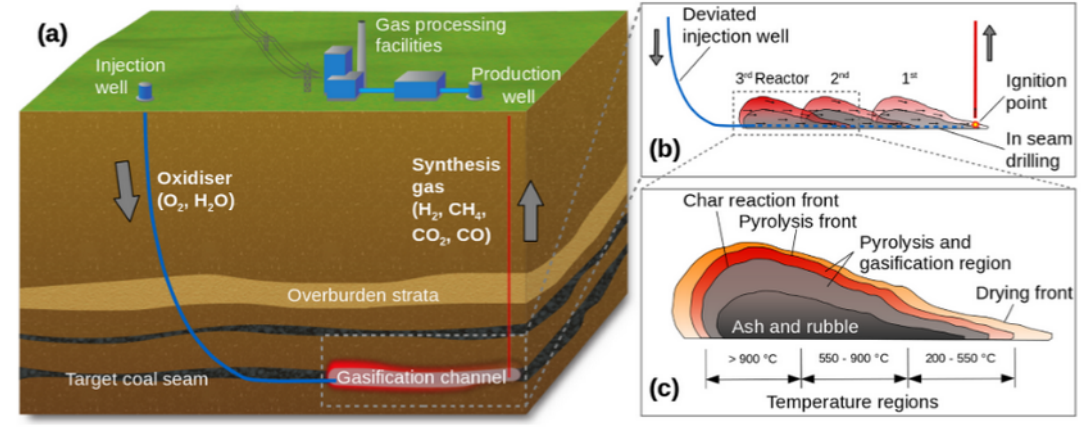
Groundwater contamination by subsurface utilisation

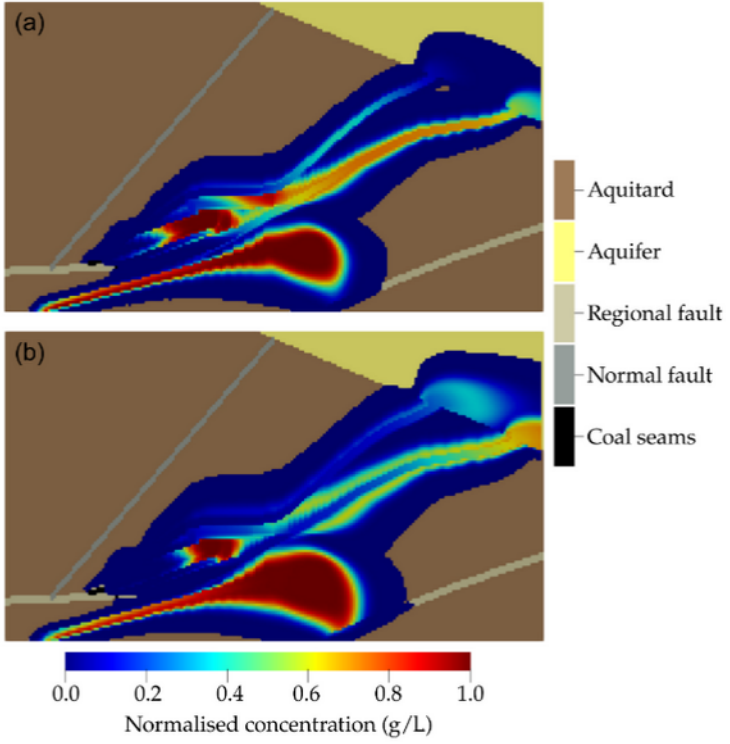
Adv. Geosci., 58, 55–66, 2022
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Numerical Analysis of Potential Contaminant Migration from Abandoned In Situ Coal Conversion Reactors

Christopher Otto[1], Svenja Steding[1], Morgan Tranter[1], Torsten Gorka[2], Mária Hámor-Vidó[3], Wioleta Basa[4], Krzysztof Kapusta[4], István Kalmár[5], and Thomas Kempka[1,6]



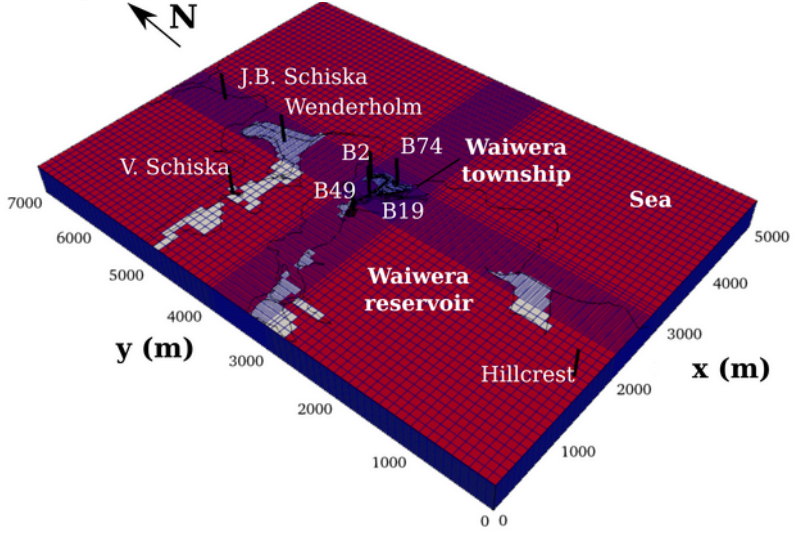


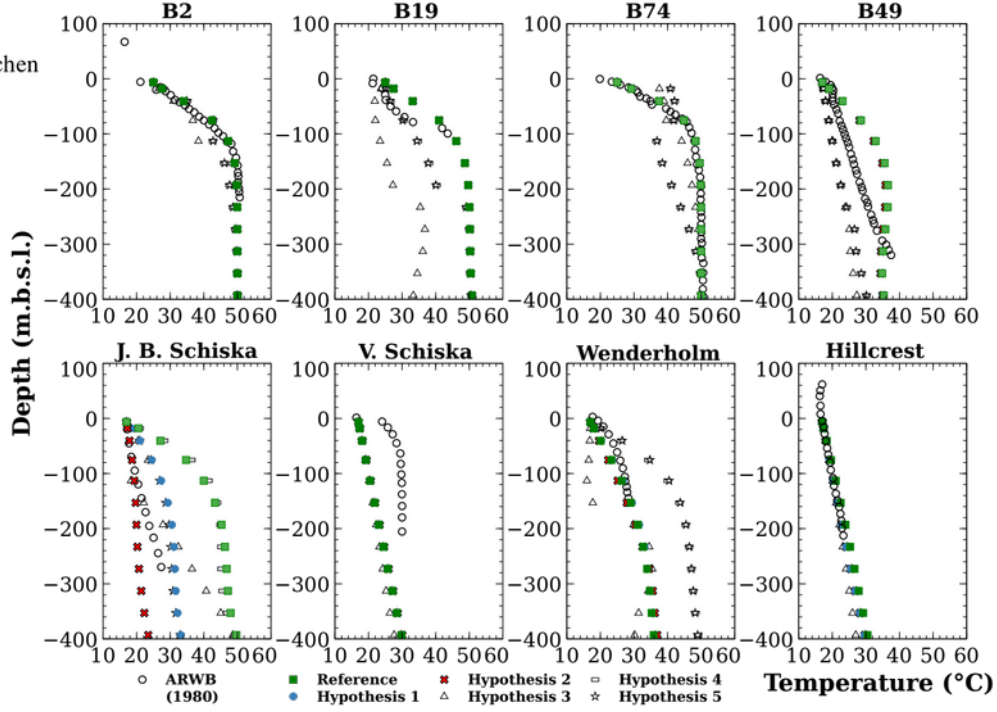
Geothermal evolution and costal groundwater salinisation

Title
Numerical simulation of spatial temperature and salinity distribution in the Waiwera geothermal reservoir, New Zealand

Titel
Numerische Simulation der räumlichen Temperatur- und Salinitätsverteilung im geothermischen Reservoir Waiwera, Neuseeland

Authors
Thomas Kempka[1,2*] and Michael Kühn[1,2]





3D permafrost formation and thawing as well as hydrate deposit development at geological time scales

Geologic Controls on the Genesis of the Arctic Permafrost and Sub-Permafrost Methane Hydrate-bearing System in the Beaufort-Mackenzie Delta

Zhen Li [1,2], Elena Chabab [1], Erik Spangenberg [1], Judith M. Schicks [1,3] and Thomas Kempka [1,2*]

